

Bentley Oakes

Assistant Professor

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🆔 0000-0001-7558-1434 🌐 bentley-oakes 📺 Bentley-Oakes

Key Academic Skills

Researcher: Lead- and co-author of dozens of publications with hundreds of citations on Google Scholar.

Collaborator: Involved in multiple international academic and industrial collaborations.

Teacher/Mentor: Extensive experience teaching courses and mentoring graduate students.

Community-Builder: Organizing and program committee member for local and global events.

Language Skills

English: Native proficiency

French: Intermediate

Education

- | | | |
|------------|---|------------------|
| PhD | McGill University , Computer Science | Montréal, Canada |
| | <ul style="list-style-type: none">Thesis: <i>A Symbolic Execution-Based Approach To Model Transformation Verification using Structural Contracts</i>Supervisors: Hans Vangheluwe and Clark Verbrugge | 2019 |
| MSc | McGill University , Computer Science | Montréal, Canada |
| | <ul style="list-style-type: none">Thesis: <i>Practical and Theoretical Issues of Evolving Behaviour Trees for a Turn-Based Game</i>Supervisor: Clark Verbrugge | 2013 |
| BSc | University of Manitoba , Computer Science (Honours) | Winnipeg, Canada |
| | <ul style="list-style-type: none">Co-op internship at Electronic Arts Inc., Montréal — Fall 2009, Summer 2010: Prototyping artificial intelligence in commercial video games.Co-op internship at Blackberry Limited (RIM), Waterloo — Winter 2009: Implementing cryptographic communication protocols. | 2011 |

Research Experience

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| Polytechnique Montréal , Assistant Professor | Montréal, Canada |
| <ul style="list-style-type: none">Department of Computer and Software Engineering (GIGL)Research topics: Accelerating the construction and deployment of digital twins through ontologies for capturing domain knowledge, and large language models and agentic AI for automating their creation. Integrating formal methods to verify digital twin correctness, and assisting users with the configuration and observability of their machine learning and verification/validation services. | Sep. 2023 – present |
| Université de Montréal , Post-Doctoral Researcher | Montréal, Canada |
| <ul style="list-style-type: none">Lab: GEODES Software Engineering Research GroupSupervisors: Houari Sahraoui and Michalis FamelisResearch topic: Assisting non-machine learning experts in constructing ML solutions by exploring tailoring of computational workflows.Research topic: Extracting and structuring developer rationale from open-source commit messages using NLP and Large Language Models, in collaboration with PhD student Mouna Dhaouadi. | Sep. 2021 – Aug. 2023 |

University of Antwerp, Post-Doctoral Researcher

- Labs: Antwerp Systems and Software Modelling, and Constrained Systems Lab
- Supervisors: Hans Vangheluwe and Joachim Denil
- Research topics: Verification and validation of cyber-physical systems, model-driven engineering, multi-paradigm modelling, co-simulation, and digital twins.

Antwerp, Belgium
Sep. 2018 – Jul. 2021

Université de Montréal, Visiting Researcher

- Host: Eugene Syriani, GEODES Software Engineering Research Group
- Research topic: Developing an interface between the AToMPM modelling tool and the ModelVerse modelling repository.

Montréal, Canada
May 2018

fortiss GmbH, Visiting Researcher

- Host: Levi Lúcio
- Research topic: Formalizing representations of model transformation languages.

Munich, Germany
Jul. – Aug. 2016

General Motors Technical Center, Visiting Researcher

- Host: Ramesh Sethu
- Research topics: Applying model transformations for code/model modernization at an industrial scale, and industrial intellectual property concerns.

Warren, USA
Oct. – Dec. 2014

Industrial Collaborations

Framework for Systematic Design of Digital Twins (DTDesign)

2019–2021

- Collaborators: Atlas Copco, Flanders Make
- Topics: Digital twin characteristics, integrating knowledge graphs and digital twins.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

Automated & Simulation-based Functional Safety Engineering Methodology (aSET)

2018–2020

- Collaborators: Dana Belgium NV, Siemens Industry Software (Leuven), Flanders Make
- Topics: Verification of safety-critical cyber-physical systems (formal methods, reinforcement learning-based fault injection for hazard discovery, simulation/visualization), DevOps for functional safety.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

Innovation in the Development of Electrical Systems For Aeronautics (INES)

2019–2020

- Collaborators: Boeing Research & Technology Europe (Madrid), Siemens Industry Software (Leuven), Flanders Make
- Topics: Co-simulation configuration, fault injection in co-simulation.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

Network for the Engineering of Complex Software-Intensive Systems for Automotive Systems (NECSIS)

2013–2016

- Collaborators: General Motors of Canada Ltd., Queen's University, University of Antwerp
- Topic: Verification of model transformations.
- Funding: \$14M from Automotive Partnership Canada and Natural Sciences and Engineering Research Council of Canada (NSERC)

Teaching Experience

Polytechnique Montréal, Course Instructor

Montréal, Canada
2024 – present

- Engineering of Digital Twins (LOG8421E) — Winter 2025, Winter 2026
- Introduction à la programmation (INF1007) — Fall 2025, Fall 2026
- Software Quality Engineering (LOG8371E) — Fall 2024
- Scientific and Technical Communication (INF6900AE/INF7900A) — Winter 2024, Fall 2024, Fall 2025, Fall 2026

Polytechnique Montréal , Guest Lecturer • XR Software Development (LOG8704) — Nov. 2024: <i>Digital Twins: Introductions and Intersections</i> • Model-Driven Software Engineering (LOG6953DE) — Nov. 2022, Nov. 2023: Model-driven engineering, usage and verification of model transformations	Montréal, Canada 2022 – 2024
McGill University , Course Lecturer/Coordinator • Foundations of Programming (COMP 202) — Winter 2015, 2017, 2018 (avg. 189 students) • Developed and presented lecture material on Java programming for students with no prior programming experience. • Created assessments and marking guides; coordinated with other instructors and TAs.	Montréal, Canada 2015 – 2018
University of Antwerp , Teaching Assistant • Model-Driven Engineering (2001WETMTR) — Fall 2020 • Modelling of Software-Intensive Systems (2001WETMSI) — Fall 2019	Antwerp, Belgium 2019 – 2020
McGill University , Teaching Assistant • Foundations of Programming (COMP 202, ×2) • Introduction to Computer Science (COMP 250, ×2) • Data Structures and Algorithms (COMP 251, ×3)	Montréal, Canada 2012 – 2014

Student Supervision

Kérian Fiter — PhD Student • Digital twin federation, trust, and observability • Ontological federation and reasoning, large language models, and time-series foundation models	2024–present
Carlos Pambo — PhD Student • Recommendations and patterns for digital twin engineering • Ontologies for digital twin engineering	2024–present
Adil Lagrou — PhD Student • Formalization of digital twin component interoperability • Ontological federation and reasoning for interoperability	2026–present
Gabrielle Gallant — Master’s Student • AI-enhanced digital twins for vibration analysis in manufacturing • Machine learning techniques including time-series foundation models • Co-supervised with Oguzhan Tuysuz	2025–present
Angelica Portocarrero — Master’s Student • Structured digital twin engineering with LLMs • Previously Mitacs Globalink Research Intern (2025)	2026–present
Fatima Aguel — Mitacs Globalink Research Intern • Ontological modelling of digital twin component repositories	Summer 2026
Mikkel Schmidt Andersen — Research Intern • Probabilistic digital twin update scheduling	2025
Louis Malassigné-Onfroy — Research Assistant • Visualization of digital twin architectures	2024
Amélie Bosphore — Research Assistant • Implementation of modelled machine learning pipelines	2024

Justin Caisse — Research Assistant

2023–2024

- Ontological modelling of machine learning pipelines

Invited Talks and Presentations

Guest lecture — Engineering Digital Twins course, McMaster University

Mar. 2026

- Host: Prof. Istvan David
- *What Does Your Digital Twin Do? A Framework and Tooling for Systematic DT Reporting*

Invited talk — SIG LLODIA (Lecture #3)

Jan. 2026

- *Towards Ontology-based Digital Twin Service Construction and Reporting*

Guest lecture — RAISE course, UCLA

Nov. 2025

- Host: Dr. Maged Elaasar
- *What is a ‘Digital Twin’ and How Do I Build One?*

Tutorial — Software Engineering for Machine Learning Applications (SEMLA)

Jun. 2025

- *What is a ‘Digital Twin’ and How Do I Build One?*

Invited talk — EDT.Community

May 2025

- *Systematic Digital Twin Reporting*
- [Video recording](#)

Invited talk — Next-Generation Cities Institute, Concordia University

May 2025

- *Accelerating Digital Twin Research and Engineering*

Conference paper presentation — ICSE 2025, Ottawa

May 2025

- *Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice* (journal-first)

Guest lecture — Engineering Digital Twins course, McMaster University

Mar. 2025

- Host: Prof. Istvan David
- *Accelerating Digital Twin Research and Engineering*

Research presentation — KARI Delegation, Polytechnique Montréal

Feb. 2025

- Advanced Air Mobility (AAM) challenges and advancements

Invited talk — Software Engineering at Montréal (SEMTL)

2025

- *Recent Work on Systematic Reporting and Construction for Digital Twins*

Guest lecture — LOG8704 XR Software Development, Polytechnique Montréal

Nov. 2024

- *Digital Twins: Introductions and Intersections*

Conference paper presentation — MODELS 2024

Sep. 2024

- *Fault localization in DSLTrans model transformations by combining symbolic execution and spectrum-based analysis* (SoSyM journal-first)

Invited talk — onto:Nexus Forum 2024

Jan. 2024

- *Accelerating Digital Twin Construction with Ontologies*
- 1st International Forum on Ontological Modeling and Analysis, NASA JPL (attended online)

Invited talk — Polytechnique Montréal

2023

- *Bridging the Gap in Verification and Validation of Complex Systems*

Guest lecture — LOG6953DE, Polytechnique Montréal Nov. 2022

- Host: Prof. Mohammad Hamdaqa
- *Model/Graph Transformations: Specification and Verification*

Invited talk — Software Engineering at Montréal (SEMTL) September seminar Sep. 2022

- *Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice*

Invited talk — Concordia University 2022

- Host: Prof. Essam Mansour
- *Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice*

Invited talk — Université du Québec à Montréal 2022

- *Capturing and Utilising Domain-Specific Knowledge*

Community Building

Workshop Organizer — onto:Nexus Workshop 2026 (International Workshop on Ontological Modeling and Analysis) 2026

- Co-located with MODELS 2026, Malaga, Spain.
- Co-organizers: Maged Elaasar (Chair), Wahab Hamou-Lhadj, Mohammad Hamdaqa, Eduard Kamburjan

Workshop Organizer — onto:Nexus Workshop 2025 (International Workshop on Ontological Modeling and Analysis) Oct. 2025

- Co-located with MODELS 2025, Grand Rapids, MI. Hybrid event with IEEE proceedings.
- Co-organizers: Maged Elaasar (Chair), Wahab Hamou-Lhadj, Mohammad Hamdaqa

Lead Organizer — Software Engineering at Montréal (SEMTL) Montréal, Canada Aug. 2022 – present

- Regular seminars for the software engineering researchers in Montréal (~40 attendees per meeting, ~30% are professors).
- Coordinating with meeting hosts on content, venue, date, and maintaining website.

Organizing Committee Member 2022–2026

- SEMLA 2026 — General Chair
- FSE 2026 — Student Volunteer Co-Chair
- MODELS 2025 — Student Volunteer Co-Chair
- ICSE 2025 — Posters Co-Chair
- SEMLA 2024, 2025 — Posters Co-Chair
- EDTconf 2024 — Publicity Co-Chair
- ANNSIM 2024 — Proceedings Co-Chair
- ANNSIM 2022, 2023 — Cyber-Physical Systems Track Co-Chair
- MODELS 2022 — Posters Co-Chair

Program Committee Member	2019–2026
• ASE Research Track — 2026 • WSC — Simulation in Digital Twins Track — 2026 • ASE NIER Track — 2025 • MODELS Artifact Evaluation Track — 2026 • MODELS — 2025 • ANNSIM — 2021–2024, 2026 • DIGITA Workshop (PerCom) — 2026 • IDETWIN Workshop (CCNC) — 2026 • International Workshop on Architecting and Engineering Digital Twins (AEDT) — 2025, 2026 • EMMSAD — 2025 • EDTconf — 2024, 2025, 2026 • MDE Intelligence — 2022, 2023, 2024 • International Workshop on Models and Evolution — 2022, 2026 • ACM Student Research Competition — 2022 • Spring/Summer Simulation Conference — 2019, 2020	
Journal Reviewer	2020–2026
• KI – Künstliche Intelligenz — 2026 • Journal of Systems & Software (JSS) — 2025 • Information and Software Technology (IST) — 2025 • Journal of Software and Systems Modeling (SoSyM) — 2020, 2021, 2023, 2025, 2026 (Top 1% Reviewer 2020, 2021; Best Reviewer Award 2025) • ACM Transactions on Software Engineering and Methodology (TOSEM) — 2024 • SIMULATION — 2023, 2024 (Best Reviewer Award June 2023) • Science of Computer Programming (SCP) — 2023 • Journal of Computer Languages (JCL) — 2022 • Journal of Object Technology (JOT) — 2022, 2026 • Empirical Software Engineering (EMSE) — 2022 • IEEE Transactions on Automation Science and Engineering (T-ASE) — 2021	
Guest Editor for Journal Special Issue	2023
• SIMULATION — <i>Modeling and Simulation for Software-Intensive Systems: from IoT to Digital Twins</i>	
Jury Member	2023–2026
• INSA Lyon / Polytechnique Montréal — PhD Thesis Jury Member (2026) • Polytechnique Montréal — PhD Thesis Jury Member (2026) • Polytechnique Montréal — Examen de synthèse (doctoral qualifying exam) President ×2 (2025–2026) • Polytechnique Montréal — Examen de synthèse (doctoral qualifying exam) Member (2026) • Polytechnique Montréal — M.Sc. Thesis Jury President ×3 (2026) • Polytechnique Montréal — M.Sc. Thesis Jury President (2024) • McGill University — PhD Thesis External Reviewer (2024) • McGill University — M.Sc. Thesis External Reviewer (2023)	
Grant Reviewer	2024–2026
• NSERC Idea to Innovation grant — 2024 • NSERC Discovery grant — 2025, 2026 • Dutch Research Council (NWO) — 2026	
Panelist	2023
• ANNSIM PhD Colloquium Panel — 2023	

Departmental Service

2024–Present

- Committee Member, Committee Rayonnement (Communications committee),
Department of Computer and Software Engineering, Polytechnique Montréal

Scholarships and Awards

SoSyM Best Reviewer Award

2025

- Journal of Software & Systems Modeling

Simulation Journal Best Reviewer Award

June 2023

SoSyM Top 1% Reviewer

2020, 2021

- Journal of Software & Systems Modeling

Best Student Paper Award

2019

- SIMULTECH — for the paper *HintCO – Hint-based configuration of co-simulations*

NSERC Postgraduate Scholarship – Doctoral (PGS D)

2015–2016

- Natural Sciences and Engineering Research Council of Canada

Lorne Trottier Science Accelerator Fellowship

2015, 2016

- McGill University

Harold H. Helm Fellowship

2013, 2014

- McGill University

Grad Excellence Award in Computer Science

2012, 2014

- McGill University

Selected Publications

Bridging the Semantic Web and Model-Based Systems Engineering with the Ontological Modeling Language

Oct. 2026

Maged Elaasar, **Bentley Oakes**, Eduard Kamburjan, Mohammad Hamdaqa, Abdelwahab Hamou-Lhadj

(International Semantic Web Conference (ISWC) — accepted)

Involving before Evolving: A Vision for Trustworthy Enterprise Digital Twin Engineering

Oct. 2026

Kérian Fiter, Adil Lagrou, **Bentley Oakes**, Franck Dervault

[10.1145/3837062.3839094](#) (MODELS Companion (EDTconf))

Model-based digital twin engineering: insights, challenges, and future directions

May 2026

Philipp Zech, Souvik Barat, Benjamin Nast, **Bentley Oakes**, Judith Michael, Steffen Zschaler, Balbir Barn, Ruth Breu

[10.1007/s10270-026-01368-8](#) (Software and Systems Modeling (SoSyM))

Automated Extraction and Analysis of Developer's Rationale in Open Source Software

2025

Mouna Dhaouadi, **Bentley Oakes**, Michalis Famelis

[10.1145/3729383](#) (Proceedings of the ACM on Software Engineering (FSE))

DTInsight: A Tool for Explicit, Interactive, and Continuous Digital Twin Reporting

2025

Kérian Fiter, Louis Malassigné-Onfroy, **Bentley Oakes**

[10.1109/MODELS-C68889.2025.00030](#) (MODELS Companion (MODELS-C))

Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State of the Practice Bentley Oakes , Michalis Famelis, Houari Sahraoui 10.1145/3638243 (ACM Transactions on Software Engineering and Methodology (TOSEM))	Apr. 2024
Toward Intelligent Generation of Tailored Graphical Concrete Syntax Meriem Ben Chaaben, Oussama Ben Sghaier, Mouna Dhaouadi, Nafisa Elrasheed, Ikram Darif, Imen Jaoua, Bentley Oakes , Eugene Syriani, Mohammad Hamdaqa 10.1145/3640310.3674085 (MODELS 2024)	2024
Towards Ontological Service-Driven Engineering of Digital Twins Bentley Oakes , Claudio Gomes, Eduard Kamburjan, Giuseppe Abbiati, Elif Ecem Bas, Sebastian Engelsgaard 10.1145/3652620.3688261 (MODELS Companion (EDTconf) — Best Short Paper Award)	2024
Toward a Systematic Reporting Framework for Digital Twins: A Cooperative Robotics Case Study Santiago Gil, Bentley Oakes , Cláudio Gomes, Mirgita Frasheri, Peter G. Larsen 10.1177/00375497241261406 (SIMULATION)	2024
openCAESAR: Balancing Agility and Rigor in Model-Based Systems Engineering Maged Elaasar, Nicolas Rouquette, David Wagner, Bentley Oakes , Abdelwahab Hamou-Lhadj, Mohammad Hamdaqa 10.1109/MODELS-C59198.2023.00051 (MODELS Companion (MODELS-C))	2023
Machine Learning-Based Fault Injection for Hazard Analysis and Risk Assessment Bentley Oakes , Mehrdad Moradi, Simon Van Mierlo, Hans Vangheluwe, Joachim Denil 10.1007/978-3-030-83903-1_12 (SAFECOMP 2021 (Computer Safety, Reliability, and Security))	2021

Publications

- See bentleyoakes.com/publications for a full list of dozens of peer-reviewed publications, including journal articles, book chapters, conference papers, workshop papers, and technical reports/theses.